Abdominal wall

Introduction to Abdomen, pelvic and perineum

The human abdomen is between the pelvis and the thorax. From the thorax (at the thoracic diaphragm) to the pelvis at the pelvic brim.

The pelvic brim stretches from the lumbosacral angle (the intervertebral disk between L5 and S1) to the pubic symphysis and is the edge of the pelvic inlet.

Functionally, the human abdomen is where most of the alimentary tract is placed.

Content: The abdomen is comprised primarily of the digestive tract and other accessory organs which assist in digestion, the urinary system, spleen, and the abdominal muscles (shown below). The majority of these organs are encased in a protective membrane termed the peritoneum. While the digestive organs and accessory organs are located within the peritoneum, the kidneys, ureters and urinary bladder are located outsider of the peritoneum, and thus, are considered by some scientists to be pelvic organs.

Many of the organs in the abdominal cavity can be **palpated** through the abdominal wall, or their position can be visualised by surface markings

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It is important to study the situation of organs and their relative position in relation to the anterior abdominal wall. The surface subdivisions have helped the descriptions of injuries and swellings to a greater extent.

The whole of the anterior abdominal wall is divided into nine regions. They are surface landmarks useful to the Medical students during their study and also Physicians and Surgeons during practice.

Important surface landmarks:

Subcostal arch; It is formed by the 7th, 8th, 9th & 10th costal cartilages.

Subcostal plane crosses the 3rd lumbar vertebra.

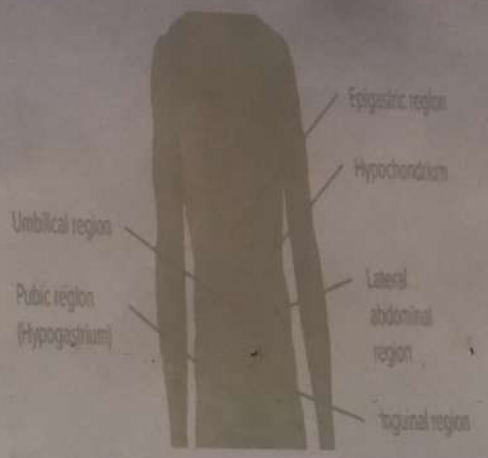
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Xipho-sternum: An effort to feel the Xiphoid process is a painful act in the living. It however lies at the 9th thoracic vert.

Linea alba: It is midline marked by a shallow vertebral groove. It is thick above the umbilicus and thin below.

Umbilicus: It is the most reliable landmark. Segmental cutaneous innervation of umbilical region is from the 10th thoracic.

Pubic crest and tubercle: They are parts of pubic bone. Pubic tubercle is felt at medial end of inguinal furrow.

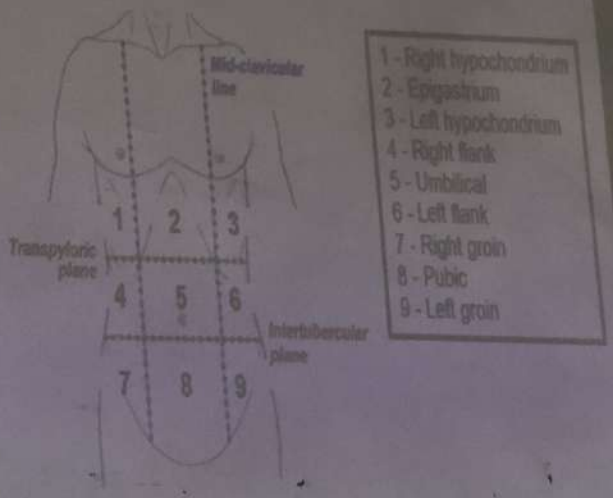


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Linea semilunaris: It is formed where the lateral border of rectus abdominis muscle produces groove line that crosses the subcostal arch at the tip of the 9th costal cartilage. The lower border of linea semilunaris reaches the pubic tubercle. At the tip of 9th costal cartage lies the fundus of gall bladder. In cholecystitis (inflammation), of gall bladder, there is tenderness at the tip of 9th costal cartilage (Murphy sign).

Inguinal ligament: It lies under the inguinal furrow and extends from the ant. Sup. iliac spine to the pubic tubercle.

Ant. Sup. Iliac Spine: It lies at the level of sacral promontory.



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Iliac crest: It is at the lateral end of inguinal furrow and corresponds to the level of 4th lumbar vertebral.

Mid-inguinal point: It is the midpoint of the line joining the anterior sup iliac spine. And the symphysis pubis.

The following plains are commonly used for division of ant. Abd. Wall.

Trans pyloric plane: It is drawn horizontally at the mid point of the line joining the suprasternal notch and symphysis. It passes through the pylorus. It also passes through the 9th costal cartilages and corresponds to the level of 1st lumbar vert.

Important structures on the plane; (i), Fundus of gall bladder, (ii), pylorus of stomach, Neck of pancreas,. (iii) Formation of portal vein. (iv). Origin of superior mesenteric artery. (v) Hilum of right kidney.

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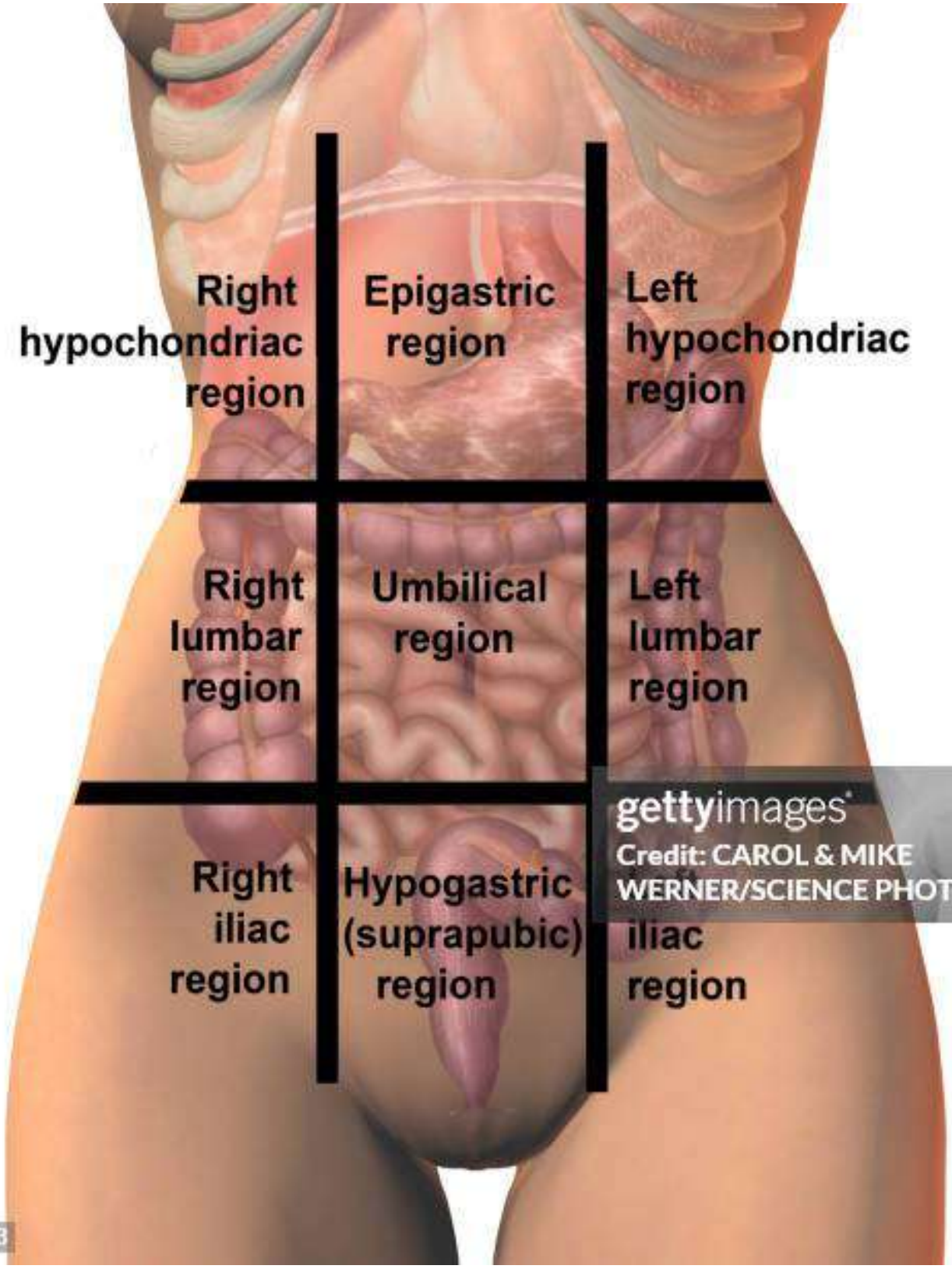
Interbubacular pain: It is formed by joining two iliac tubercles. It is at the level of 5th lumbar vertebral.

The right vertical plane is drawn from the right mid-inguinal point.

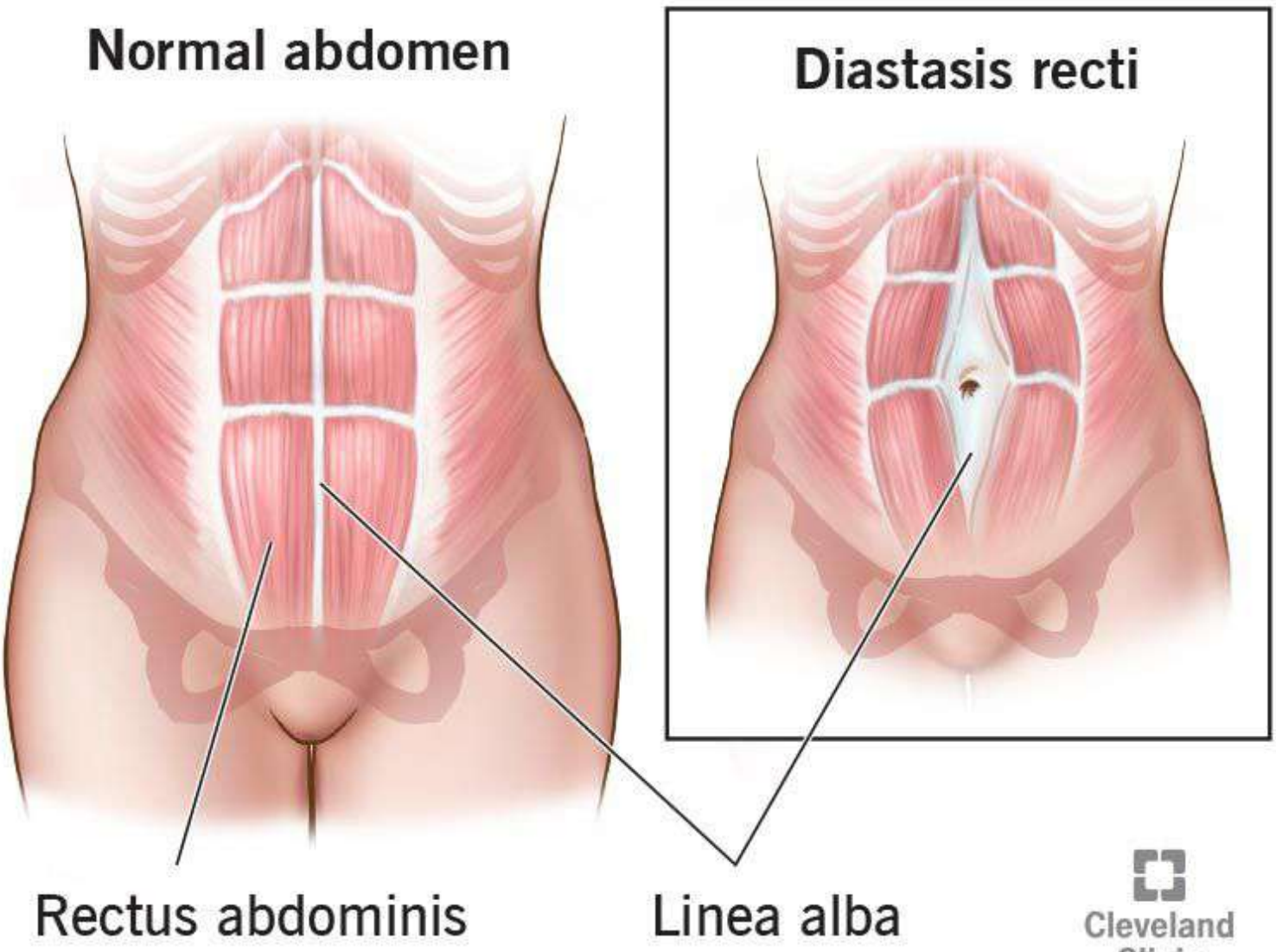
The left vertical plane is drawn from the left inguinal point.

The transpyloric, intertubercular, the right and left vertical planes are marked on the Anterior Abdominal. Wall for the purpose of 9 divisions.

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Right hypochondriac region
Epigastric region
Left hypochondriac region
Right lumbar region
Umbilical region
Left lumbar region
Right iliac region
Hypogastric (suprapubic) region
iliac region
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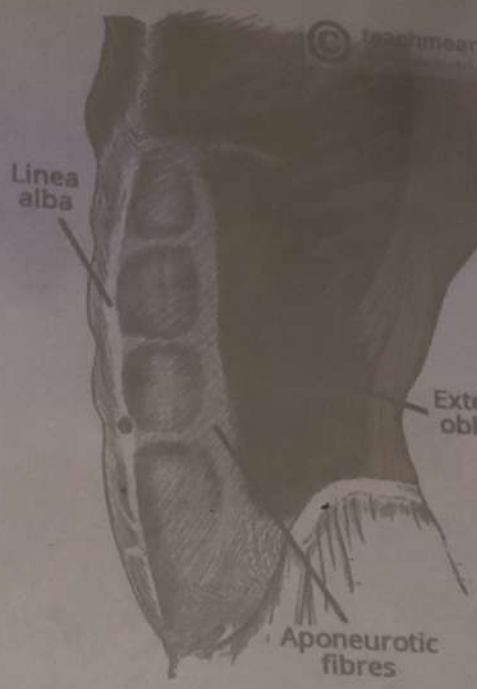
Linea Alba



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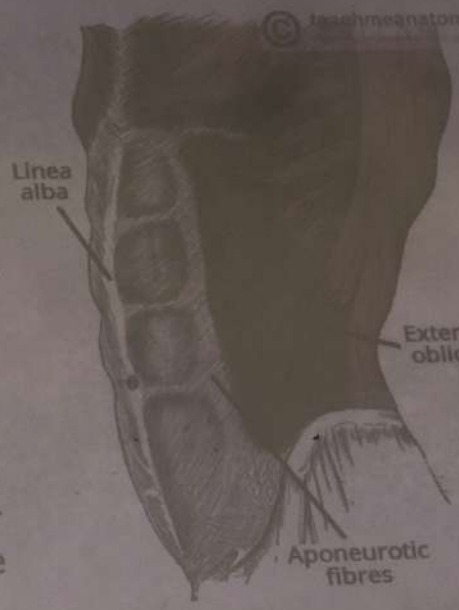
Anterior abdominal wall muscles: Anterior abdominal wall comprises of three flank muscles; External, internal and transversus abdominis muscles and one pair of straight muscle, rectus abdominis and a pair of small muscles- pyramidalis.

External oblique: A paired flat muscle. It originates by fleshy slips middle of shaft lower eight ribs fibres run downward, forward and medially. Upper four fibres interdigitate with serratus anterior. Lower four fibres interdigitate with costal fibres of latissimus dorsi.



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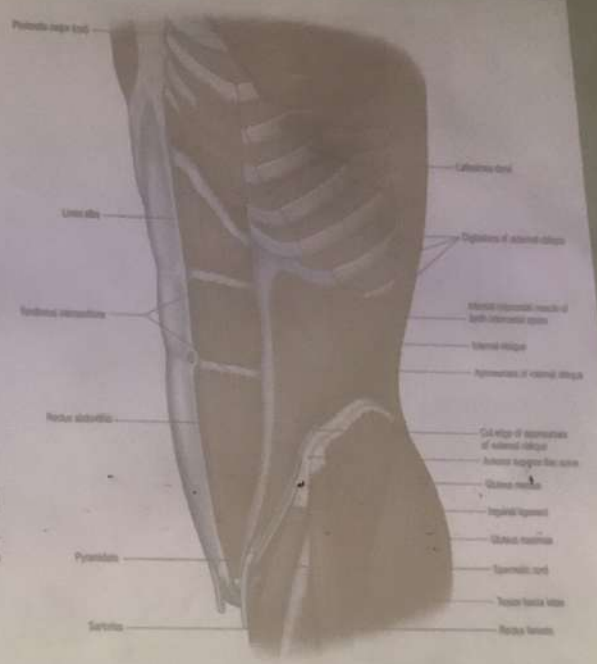
Insertion- It has very wide insertion mostly aponeurotic into xiphoid process, linea alba, pubic symphysis, pubic crest, pectinate line of pubis. It has three borders, the free upper border extends horizontally from the 5th rib to xiphoid sternum. It is the only structure that forms anterior sheath of rectus muscle. The free posterior border is fleshy and forms the anterior border of triangle of Petit. The lower border extends from the anterior superior iliac spine to pubic tubercle forming inguinal ligament.



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Triangle of Petit: The triangle has as its anterior border- fleshy posterior part of external oblique muscle, posterior border- latissimus dorsi, base inferiorly by the iliac crest and floor by the internal oblique. It is a site for rare lumbar hernia.

Internal oblique: It arises via fleshy fibres from the whole length of lumbar fascia, anterior 2/3 of the intermediate area of iliac crest, lateral 2/3 of inguinal ligament. From their origin, fibres run upward, forward and medially crossing external oblique at right angles.



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Insertion: From lumbar fascia the muscle fibres run upwards along costal margins and become aponeurotic at the tip of 9th costal cartilage. Below the cartilage, the aponeurotic fibres split lateral of rectus muscle with the two layers re-joining at linear alba. The tendinous intersection of rectus muscle adheres to anterior layer. About 2.5cm below the umbilicus, the posterior layer end in curved free margin, concave downward called arcuate line or semi-circular line (Douglas). Below that point the aponeurotic passes wholly in front of rectus muscle to linear alba. Muscle fibres from inguinal ligament continue into aponeurosis that inserts from crest of pubic bone to pectinate line more laterally. Innervation Lower 6 intercostal.

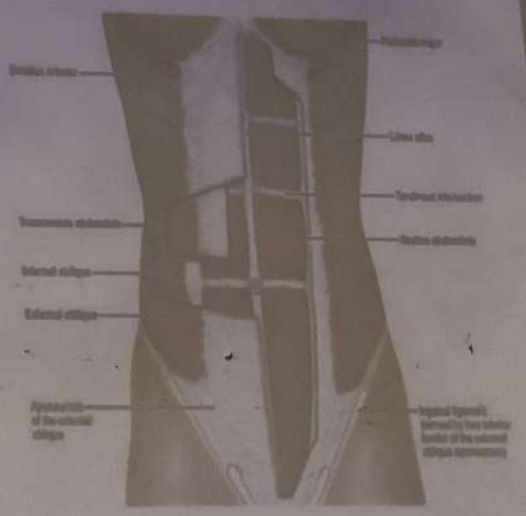
Transversus abdominis: It has a very long origin from costal margin to inguinal ligament. Beginning from the internal surface of lower six costal cartilages interdigitating with costal origin of diaphragm,

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also from lumbar fascia lateral to quadratus lumborum, from anterior two-third of internal lip of iliac crest, from fascia over the iliacus and lateral ½ of inguinal ligament.

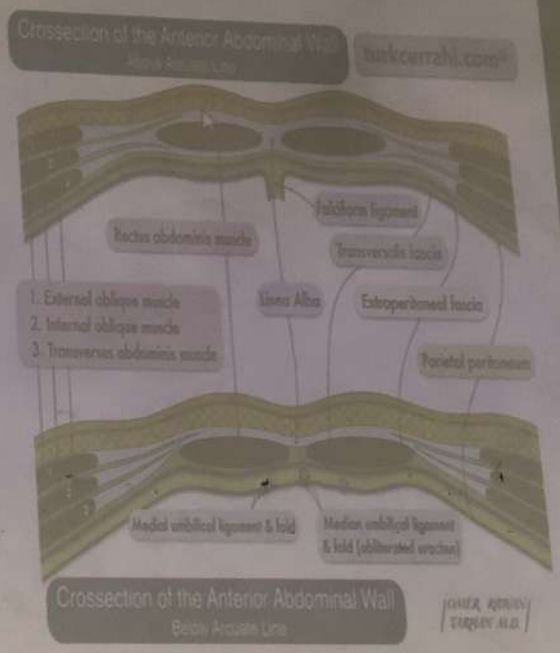
Insertion: Muscle fibres become aponeurotic and passes behind rectus abdominis to fuse with internal oblique into xiphoid sternum, linea alba. Below arcuate line, the aponeurosis passes wholly in front of rectus abdominis behind the aponeurosis of internal oblique which it accompanies laterally as conjoint tendon inserting on pubic crest to the pectinate line.

Anterior abdominal wall



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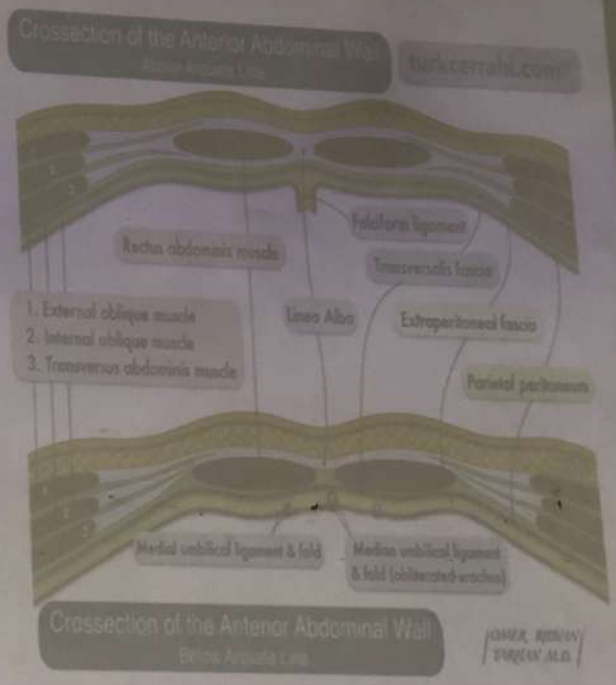
Rectus sheath: The aponeurosis of internal oblique splits into anterior and posterior layers to enclose rectus muscle. The external oblique aponeurosis fuses with the anterior layer forming anterior layer of the sheath. The transversus aponeurosis with the posterior layer forming the posterior layer of the sheath. A little below the umbilicus, the arrangement ceases. Below that point, the three aponeurosis pass in front of the muscle. There is a free lower concave margin, the arcuate line.



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Below the umbilicus of the internal and transversus abdominis fuse completely forming the conjoint tendon. Posterior layer of the sheath attaches at the costal margin. However, rectus abdominis passes above the costal margin and extends up to the 5th costal cartilage in which case external oblique abdominis is the only structure overlying the muscle.

Rectus abdominis: It has two heads, medial head is from the anterior pubic ligament while lateral head is from pubic crest. From the origins,

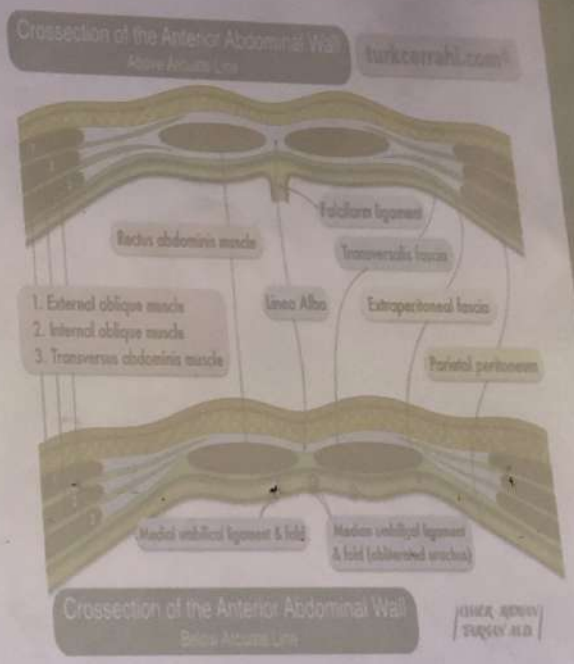


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The muscle belly rapidly thickens. The muscle from each side of the midline lies initially edge to edge but upward they broadens and crosses the rib cage up to the 5th costal cartilage for insertion. It has three intersections. One at the level of xiphoid process, one at the level of umbilicus and another between the two.

Innervation: The lower six intercostal nerves with the inclusion of the 1st lumbar.

Pyramidalis: It is a triangular muscle from the anterior surface of the pubis. The fibres then passes upward and medially to the linea alba.



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Clinical Anatomy: The multi-layered muscles have important functions like protecting organs and assisting in bodily movements.

Its landmarks structures like linea alba, umbilicus and inguinal canal are crucial for surgical procedures.

Its multi-layered is important in understanding abdominal hernia. Weakness of abdominal wall caused by trauma, surgery, or chronic strain can lead to the protraction of internal organ through the muscle.

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